

Shambhavi Mishra

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PROFESSIONAL EXPERIENCE

Centre for Railway Information Systems *Web Development Trainee* 15 Jun 2026 – 31 Jul 2026 | Delhi
Web Development Trainee | Centre for Railway Information Systems

- Developed a real-time web application using **JavaScript, Socket.io** and **Redis Pub/Sub**, supporting horizontally scalable messaging with sub-50ms delivery.
- Implemented **backend services, REST APIs and Redis caching**, reducing MongoDB reads by ~80% through optimized data access and 24-hour TTL caching.
- Developed and **debugged** real-time SOS/location services using GPS, camera recording and Leaflet.js; secured APIs using JWT, bcrypt and role-based middleware.

PROJECTS

- TraceX (Agentic Merchant Risk Detection System) | Python, React, FastAPI, XGBoost, LangGraph, Groq API, ChromaDB, SHAP, SQLAlchemy**
- Built a fraud-spike detection pipeline using **Isolation Forest + XGBoost**, achieving **0.985 PR-AUC** on unseen merchants vs. **0.785 Robust Z-Score** and **0.765 Isolation Forest** baselines.
 - Optimized the detection threshold using false-positive/false-negative costs, achieving **99.6% recall, 66.5% precision**, and only **1 false negative across 235 positive cases**.
 - Built a **LangGraph** multi-agent workflow with Policy-RAG, Evidence, and Decision agents for contextual risk investigation, with tool-based reasoning and automated workflow validation.
- ResumeAlign AI (Agentic AI Resume Analyzer) | React, Node.js, Express.js, MongoDB, Groq API, Llama 3.x, Tailwind CSS, JWT**
- Built ATS scoring across **5 weighted components**, analyzing 40 technical keywords, 26 action verbs, and 8 resume sections.
 - Achieved ~31 ms median latency across 5 test runs with **automated resume validation and Selenium-based UI testing**.
 - Built an **agentic workflow** using **Groq LLaMA 3**, tool calling and SSE streaming for context-aware iterative analysis, self-correction and response generation.
- Shiksha Setu (Offline-First Digital Learning Platform) | React Native, TypeScript, JavaScript, Node.js, Express.js, PostgreSQL, SQLite, Prisma, JWT**
- Built a full-stack application with **REST APIs, PostgreSQL and SQLite** for offline course, quiz and progress management; **designed and executed test cases** for API endpoints and data synchronization.
 - Implemented **data processing and synchronization** using queued progress and quiz events, validating offline/online workflows and edge cases through **automated testing**.
 - Developed **role-based access control** for students, teachers and admins, with database-backed course authoring and progress tracking.
- Finsight (Personal Finance Dashboard) | React, JavaScript, Node.js, Express, MongoDB, Recharts, Zustand, JWT**
- Built a responsive React frontend using **reusable components and client-server architecture, a consistent design system**, interactive Recharts visualizations, and JWT authentication.
 - Implemented Zustand-based state management for financial data, budget tracking, category alerts, and dynamic dashboard updates.
 - Developed responsive layouts, animations, dark/light mode, and interactive financial insights for a consistent user experience across screen sizes.
- Student Performance Predictor | Python, Scikit-learn, Flask, Pandas, Random Forest**
- Architected an end-to-end **ML pipeline using Random Forest Regressor** achieving **92%+ R² accuracy** to predict student final scores from five academic features across **400 student** records.
 - Built a Flask web interface enabling real-time student performance prediction.

SKILLS

Languages	Frameworks & Libraries	Backend & Databases	Developer Tools
JavaScript, TypeScript, Python, Java, C++, HTML5, CSS3, SQL	React.js, React Native, Tailwind CSS, Socket.io, Mongoose, Redis, API Integration	Node.js, Express.js, REST APIs, FastAPI, JSON, JWT, MongoDB, MySQL, SQLite, PostgreSQL	Git, GitHub, Jenkins, Selenium, CI/CD, Postman, Thunder Client, AWS, Docker, Test Automation
Core Competencies	AI/ML & LLM		
DSA (500+), OOP, Operating Systems, DBMS, Computer Networks, System Design, Backend Development, Software Development, API Integration, Debugging, SDLC, Database Concepts, API Testing, Debugging, Troubleshooting	Generative AI, LLM Applications, LLM Evaluation, RAG, Prompt Engineering, LangGraph, LangChain, OpenAI API, Groq API, Claude API, ChromaDB, Sentence Transformers, XGBoost, SHAP, Scikit-learn, Pandas, NumPy, Data Preprocessing		

EDUCATION

B.Tech. in Computer Science & Engineering, *Bhagwan Parshuram Institute of Technology, Rohini, Delhi*

- CGPA - 8.41

2023 – 2027 | Delhi

ACHIEVEMENTS

- Solved 500+ DSA problems in Java on LeetCode.
- Participated in Smart India Hackathon (SIH) 2025, Finalist in SholkaDecode (NSUT) and BuildWars (MAIT)

COURSES

Samsung Innovation Campus(Artificial Intelligence) *📍, Samsung*

12/2025 – 03/2026